

REGION OF WATERLOO INTL, ON, Canada

WMO: 713680

Lat: 43.4616N

Lon: 80.3856W

Elev: 1055

StdP: 14.14

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) 0.2	(d) -12.2	(e) 3.0	(f) -4.1	(g) -6.4	(h) 4.1	(i) 1.1	(j) 29.2	(k) 21.8	(l) 26.8	(m) 20.5	(n) 5.6	(o) 310	(p) 0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.8	(c) 87.0	(d) 71.2	(e) 84.1	(f) 70.0	(g) 81.4	(h) 68.7	(i) 74.0	(j) 82.9	(k) 72.4	(l) 80.6	(m) 70.8	(n) 78.1	(o) 11.6	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	120.5	78.5	69.8	113.9	76.6	68.2	107.9	75.1	38.3	82.8	36.8	80.8	35.4	78.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.4	(b) 22.4	(c) 19.9	(e) -13.7	(f) 91.1	(g) 7.9	(h) 3.0	(i) -19.4	(j) 93.2	(k) -24.1	(l) 95.0	(m) -28.6	(n) 96.7	(o) -34.3	(p) 98.9		
(4) 25.4	(5) 22.4	(6) 19.9	(e) -14.1	(f) 77.4	(g) 7.9	(h) 2.1	(i) -19.8	(j) 78.9	(k) -24.4	(l) 80.1	(m) -28.8	(n) 81.2	(o) -34.6	(p) 82.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	20.4	21.0	30.4	42.9	55.3	64.1	68.6	66.6	60.3	48.1	37.0	27.3		
	DBStd	19.12	11.65	10.83	11.15	8.72	8.37	6.55	5.52	5.34	6.95	8.18	8.40	8.85		
	HDD50	3863	918	811	613	244	38	0	0	0	4	137	394	704		
	HDD65	7541	1383	1231	1072	662	321	94	25	45	175	525	839	1169		
	CDD50	2155	0	0	7	32	203	422	578	516	313	79	5	0		
	CDD65	358	0	0	0	0	21	66	139	96	34	2	0	0		
	CDH74	3493	0	0	4	12	255	676	1352	821	345	28	0	0		
	CDH80	872	0	0	0	0	51	191	378	172	77	3	0	0		
(14) Wind	WSAvg	9.0	10.8	10.6	10.1	10.6	8.7	7.8	7.0	6.7	7.0	8.7	9.6	10.6		
(15) Precipitation	PrecAvg	33.5	2.9	2.3	2.1	2.9	3.2	3.1	3.5	2.6	2.9	3.0	2.8	2.3		
	PrecMax	42.2	4.7	4.4	5.2	5.1	6.1	6.7	6.8	5.1	6.1	5.4	5.7	4.4		
	PrecMin	22.7	1.3	1.1	0.4	0.9	0.8	1.1	0.6	0.3	0.8	0.7	0.6	0.6		
	PrecStd	5.0	1.0	0.9	1.2	1.2	1.3	1.4	1.9	1.2	1.5	1.3	1.2	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	53.0	68.4	75.8	84.9	88.8	91.2	88.9	86.7	77.7	63.5	52.1		
		MCWB	50.9	49.5	59.8	60.4	68.2	71.9	73.1	72.3	72.7	65.8	55.8	50.2		
	2%	DB	45.4	44.7	57.6	69.4	80.8	85.5	87.5	84.3	82.0	71.7	58.0	46.7		
		MCWB	43.5	41.8	50.6	56.5	66.9	70.2	71.4	70.6	69.2	62.3	52.7	44.7		
	5%	DB	39.7	38.7	51.7	64.3	76.3	81.5	84.2	81.4	78.1	66.6	54.2	42.3		
		MCWB	37.9	35.9	46.8	52.7	64.2	68.6	70.2	69.5	67.1	59.7	49.7	40.4		
	10%	DB	35.7	35.0	46.2	59.0	71.7	77.3	81.2	78.8	73.9	62.0	50.2	38.4		
		MCWB	34.1	32.8	41.7	49.7	60.7	66.0	68.3	67.9	64.8	57.1	46.8	36.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	51.1	60.4	61.7	71.6	74.5	76.4	76.3	74.5	68.8	57.8	50.9		
		MCDB	51.3	52.5	67.9	70.4	81.3	84.5	86.1	81.7	83.4	74.7	60.7	52.2		
	2%	WB	43.2	42.2	52.2	58.7	68.5	72.4	74.2	73.3	71.7	64.1	53.7	44.9		
		MCDB	44.3	44.9	55.5	66.9	78.3	82.2	83.1	81.2	79.2	68.4	56.6	46.1		
	5%	WB	37.8	36.0	47.1	55.2	65.7	70.2	72.4	71.4	69.1	60.7	50.1	40.6		
		MCDB	39.4	38.3	51.3	61.8	73.6	78.2	80.5	78.0	74.8	66.0	53.5	42.2		
	10%	WB	34.1	33.0	41.9	50.7	62.7	68.0	70.7	69.7	66.6	57.6	47.0	36.8		
		MCDB	35.4	34.8	45.7	57.5	69.4	75.2	78.0	76.0	72.1	61.1	50.2	38.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	15.1	16.4	19.8	21.3	21.1	21.8	21.3	21.5	17.7	14.4	11.2		
		MCDBR	15.0	18.6	25.2	29.7	27.0	25.8	25.1	24.2	26.6	25.3	21.8	15.0		
	5% WB	MCWBR	14.5	15.9	18.1	18.7	15.1	13.4	12.0	12.3	14.5	16.4	16.7	13.8		
		MCWBR	14.8	17.2	23.5	25.7	24.1	22.0	21.4	19.5	21.7	22.6	19.3	14.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.286	0.332	0.326	0.363	0.387	0.419	0.411	0.399	0.368	0.351	0.305	0.286		
		taud	2.371	2.188	2.391	2.333	2.309	2.235	2.277	2.298	2.377	2.412	2.505	2.438		
	Ebn at Noon	Ebn at Noon	268	269	290	286	281	271	272	271	272	262	261	259		
		Edh at Noon	25	36	33	38	40	43	41	39	33	29	22	21		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	446	709	1111	1409	1708	1885	1875	1638	1314	787	486	325		
	RadStd	RadStd	56	87	114	160	166	127	128	83	126	75	66	42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page